torch.stack#

https://pytorch.org/docs/stable/generated/torch.stack.html#torch.stack

Description:

Concatenates a sequence of tensors along a new dimension. All tensors need to be of the same size. torch.cat() concatenates the given sequence along an existing dimension. tensors (sequence of Tensors) – sequence of tensors to concatenate

Function Signature

```
torch.stack(tensors, dim=0, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

`Keyword`

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> x = torch.randn(2, 3)
>>> x
tensor([[ 0.3367,  0.1288,  0.2345],
        [ 0.2303, -1.1229, -0.1863]])
>>> torch.stack((x, x)) # same as torch.stack((x, x), dim=0)
tensor([[[ 0.3367,  0.1288,  0.2345],
         [ 0.2303, -1.1229, -0.1863]],

        [[ 0.3367,  0.1288,  0.2345],
         [ 0.2303, -1.1229, -0.1863]]])
>>> torch.stack((x, x)).size()
torch.Size([2, 2, 3])
>>> torch.stack((x, x), dim=1)
tensor([[[ 0.3367,  0.1288,  0.2345],
         [ 0.3367,  0.1288,  0.2345]],

        [[ 0.2303, -1.1229, -0.1863],
         [ 0.2303, -1.1229, -0.1863]]])
>>> torch.stack((x, x), dim=2)
tensor([[[ 0.3367,  0.3367],
         [ 0.1288,  0.1288],
         [ 0.2345,  0.2345]],

        [[ 0.2303,  0.2303],
         [-1.1229, -1.1229],
         [-0.1863, -0.1863]]])
>>> torch.stack((x, x), dim=-1)
tensor([[[ 0.3367,  0.3367],
         [ 0.1288,  0.1288],
         [ 0.2345,  0.2345]],

        [[ 0.2303,  0.2303],
         [-1.1229, -1.1229],
         [-0.1863, -0.1863]]])
```

Notes & Warnings

> **NOTE** See also torch.cat() concatenates the given sequence along an existing dimension.

Detailed Documentation

Documentation

Concatenates a sequence of tensors along a new dimension.

All tensors need to be of the same size.

torch.cat() concatenates the given sequence along an existing dimension.

tensors (sequence of Tensors) – sequence of tensors to concatenate

dim (int, optional) – dimension to insert. Has to be between 0 and the number of dimensions of concatenated tensors (inclusive). Default: 0

out (Tensor, optional) – the output tensor.

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